FEBUXOSTAT

CV death risk

Second-line to allopurinol

CHEMICAL NAME 2-[3-cyano-4-(2-methylpropoxy)phenyl]-4-methyl-1,3-thiazole-5-carboxylic acid

CLASS Non-purine selective xanthine oxidase inhibitor

TRADE NAMES Adenuric · Uloric · Feburic

THERAPEUTIC USES / INDICATIONS

- Chronic hyperuricaemia with gout (urate deposits, tophi) when allopurinol is not tolerated or contraindicated.
- Prevention of tumour lysis hyperuricaemia (120 mg dose).

ADMINISTRATION

- **PO** once daily with or without food.

DOSAGES

- **Gout:** 80 mg once daily; increase to 120 mg if urate >360 µmol/L (6 mg/dL) after 2–4 weeks.
- **Tumour lysis:** 120 mg once daily, starting 2 days before chemotherapy.
- Mild–moderate renal impairment: no adjustment.
- Give flare prophylaxis (colchicine/NSAID) for at least 6 months.

PHARMACOKINETICS

- **Peak:** 1–1.5 h. Urate lowering within days.
- Metabolised in liver (glucuronidation, oxidation).
- Excreted in urine and faeces. **t½:** 5–8 h.

PHARMACODYNAMICS

- Selectively inhibits **xanthine oxidase** (both oxidised and reduced forms).
- Reduces uric acid production; does not affect other purine enzymes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, headache.
- Rash.
- Gout flares at start.

ADVERSE EFFECTS (SERIOUS)

- Cardiovascular death in patients with established CV disease.
- Liver function abnormalities, hepatitis.
- Severe hypersensitivity (SJS, DRESS).
- Thyroid changes (↑ TSH).

CONTRAINDICATIONS & CAUTIONS

- Use with azathioprine or mercaptopurine.
- Ischaemic heart disease or heart failure (avoid unless no alternative).
- Starting during acute attack.
- Severe hepatic impairment.

KEY INTERACTIONS

- **Azathioprine, mercaptopurine, theophylline:** ↑ toxicity — avoid/monitor.

NURSING MANAGEMENT

- Check cardiac history before starting.
- Monitor LFTs before and periodically; check TSH.
- Monitor serum urate to target.
- Watch for chest pain, breathlessness, rash.
- Make sure flare prophylaxis is prescribed.

PATIENT EDUCATION

- Take every day to prevent attacks.
- Report chest pain, palpitations, rash or yellow skin.
- Early flares do not mean the drug is failing.
- Limit alcohol and purine-rich foods.

KEY POINTS

- Selective xanthine oxidase inhibitor.
- Avoid with azathioprine/mercaptopurine.
- Alternative when allopurinol unsuitable.
- Caution in heart disease.

SOURCES CHECKED **[54]** Febuxostat 120 mg Film-coated Tablets (UK SmPC) **[55]** Febuxostat, oral (US label) **[56]** Febuxostat, oral (US label, Novadoz Pharmaceuticals LLC)

Full references p. 432–457